Word embeddings and scaling to big data

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In these papers and Google code archive, you will learn more about methods and software to learn word embeddings (numerical vector representations of word meanings) from corpora, and the impacts of scaling from small data to large real-world data-sets.

- E Atwell. 1986. A parsing expert system which learns from corpus analysis atwell86wordEmbeddings.pdf
- M Banko and E Brill. 2001. Scaling to very very large corpora for natural language disambiguation banko01largeCorpora.pdf
- T Mikolov et al. 2013. Efficient Estimation of Word Representations in Vector Space mikolov13word2vec.pdf
- Google code archive – word2vec code.google.com/archive/p/word2vec/

AI Research papers

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AI research is reported in a paper presented at a conference and then published in a conference **proceedings** (or **journal**)

e.g. ACL: Association for Computational Linguistics
https://aclweb.org

Lists of computational linguistics journals, conferences

journals https://aclweb.org/aclwiki/Journals

conferences https://www.aclweb.org/portal/events

Research papers: ACL Anthology https://aclanthology.org/

note: Latin plural - 1 corpus, 2+ corpora (corpuses)

note: is “Proceedings” singular or plural ?? (check in SkE)

Why read AI research papers?

To learn about AI research …

Also, ideas for a research proposal?

AI research paper is similar to a research proposal:

Research aims & objectives

Background – related research

Methods – Programme of research work (usually no durations)

Conclusions - contribution to knowledge, importance

PLUS: **Results** - not usually in a research proposal

(but proposal can add a “pilot study” for “proof of concept”)

E Atwell. 1986. A parsing expert system which learns from corpus analysis

Aim: a discovery procedure for grammar, given a corpus

Background: English grammar, PoS taggers, Markov bi-grams

Methods: frequent words: list of left-contexts and right-contexts

“all possible pairings of one word with another word (w1, w2) were compared; if the context-lists of w1 were significantly similar to the context-lists of w2, the two were deemed to belong to the same word-class”

Conclusions: 1st corpus linguist to learn word embeddings from a corpus: numerical vector representations of word meanings

BUT: “200,000-word corpus … only 175 words occur 100 times … several weeks on ICL-2960 university mainframe computer”

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M Banko and E Brill. 2001.
Scaling to very very large corpora for natural language disambiguation

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Aim: "evaluate the performance of different learning methods … when trained on orders of magnitude more labeled data"

Background: ML classifiers "Confusion set disambiguation ... {weather,whether} … labeled training data is essentially free"

Method: evaluate range of classifiers on range of data sizes

Conclusions: More data -> higher accuracy, for all classifiers

"… reconsider the trade-off between spending time and money on algorithm development versus spending it on corpus development"

Scaling to very very large corpora for natural language disambiguation

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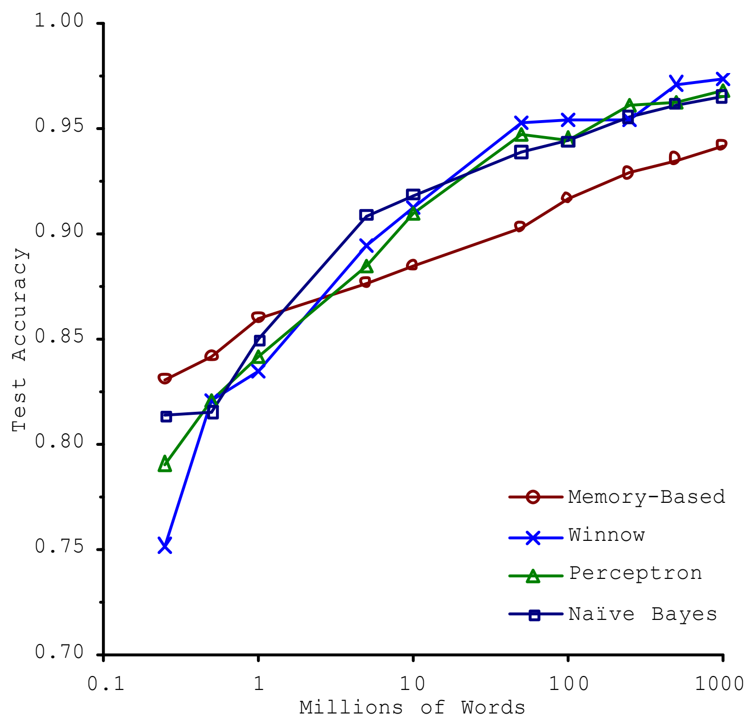

Figure 1. Learning Curves for Confusion Set Disambiguation

More data→ higher score

Data: 200K … 1000M

Accuracy: 75% … 97%

… for all classifiers

Winnow: worst to best

Case-based: best to worst

… so, "best" classifier depends on training set size NOT "best" ML algorithm

Scaling to very very large corpora for natural language disambiguation

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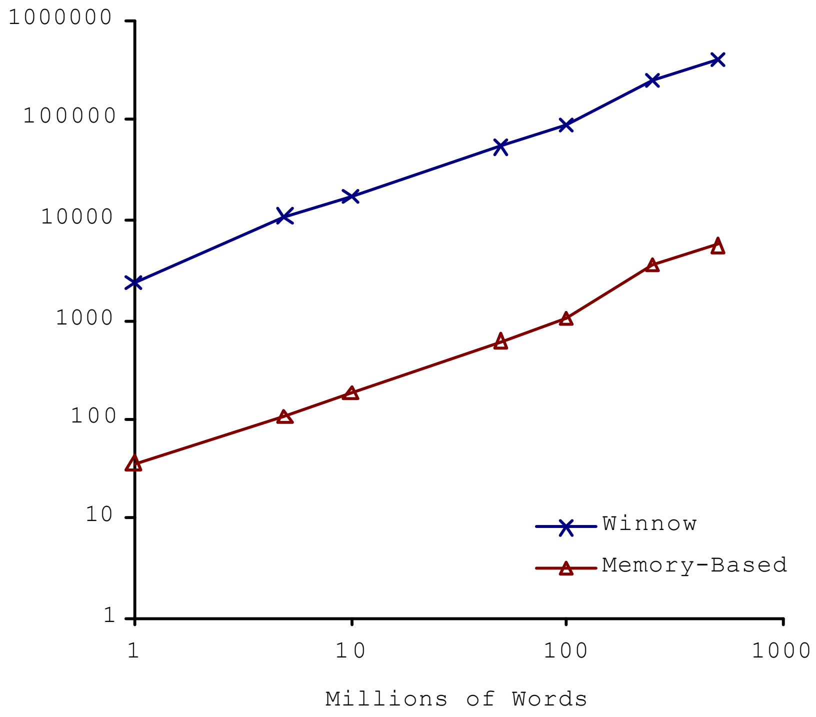

Figure 2. Representation Size vs. Training Corpus Size

Labelled training data is "free" … but other costs:

ML model size grows proportional to data size

Processor, memory grows

(Maybe no problem for Microsoft etc but others?)

Scaling to very very large corpora for natural language disambiguation

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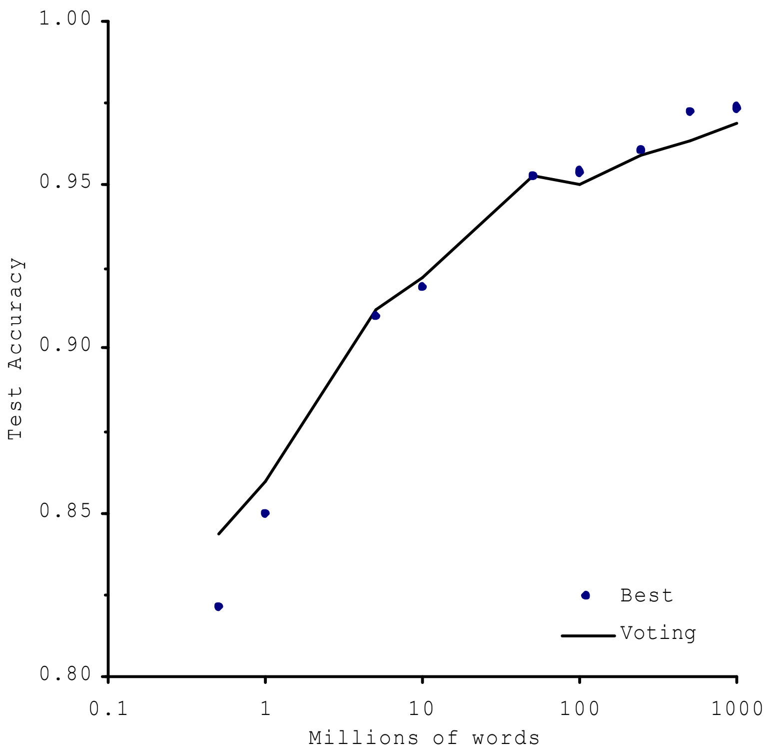

Figure 3. Voting Among Classifiers

Ensemble: several ML classifiers vote

Small data: voting > best

Big data: best > voting

(Ensemble is generally better unless you have very large training set)

Scaling to very very large corpora for natural language disambiguation

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When annotated data is not free:

Active Learning: "… choose the most uncertain instances for manual annotation"

Semi-Supervised Learning: "… choose the instances that have the highest probability of being correct for automatic labelling"

"… direct efforts towards increasing the size of annotated training collections, while deemphasizing the focus on comparing different learning techniques trained only on small training corpora."

T Mikolov et al. 2013. Efficient Estimation of Word Representations in Vector Space

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Aims: “novel model architectures for computing continuous vector representations of words from very large data sets ... and … a new comprehensive test set for measuring both syntactic and semantic regularities”

Background: vector representations in neural network i/o, Feedforward Neural Network Language Model (NNLM)

Training complexity is high (processor, memory, duration)

Methods: Continuous Bag-of-Words, Continuous Skip-gram fast training; new Word Relationships test set for evaluation

Conclusions: CBOW and Skip-gram have much lower computational complexity than feedforward and recurrent NNs

Comprehensive test set will help the research community

Efficient Estimation of Word Representations in Vector Space

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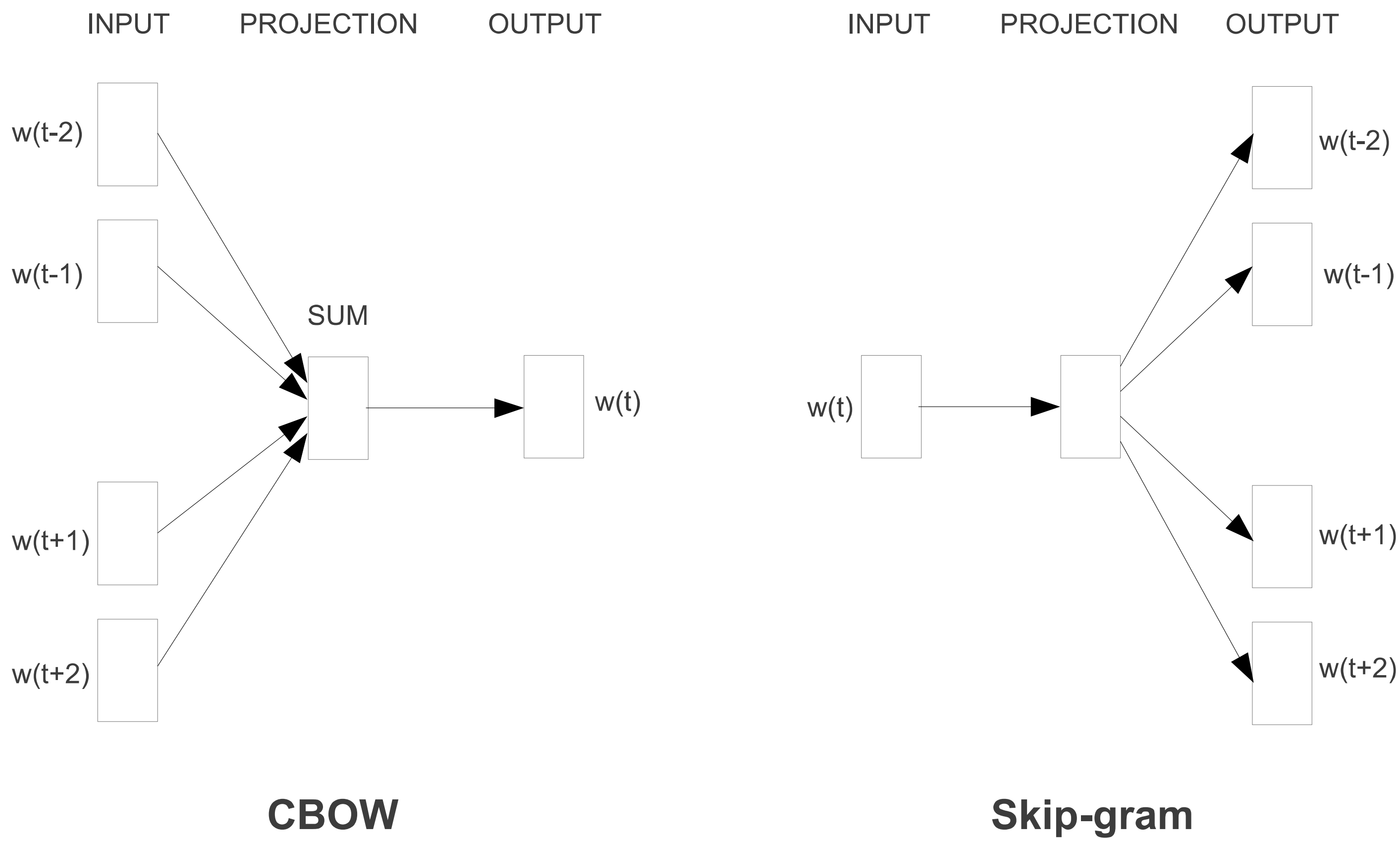

Figure 1: New model architectures. The CBOW architecture predicts the current word based on the context, and the Skip-gram predicts surrounding words given the current word.

Efficient Estimation of Word Representations in Vector Space

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“To compare the quality of different versions of word vectors, previous papers typically use a table showing example words and their most similar words, and understand them intuitively. … we define a comprehensive test set that contains five types of semantic questions, and nine types of syntactic questions. Two examples from each category are shown in Table 1. Overall, there are 8869 semantic and 10675 syntactic questions. … Question is assumed to be correctly answered only if the closest word to the vector computed using the above method is exactly the same as the correct word in the question; synonyms are thus counted as mistakes.”

Table 1: *Examples of five types of semantic and nine types of syntactic questions in the Semantic-Syntactic Word Relationship test set.*

Type of relationship	Word Pair 1		Word Pair 2	
Common capital city	Athens	Greece	Oslo	Norway
All capital cities	Astana	Kazakhstan	Harare	Zimbabwe
Currency	Angola	kwanza	Iran	rial
City-in-state	Chicago	Illinois	Stockton	California
Man-Woman	brother	sister	grandson	granddaughter
Adjective to adverb	apparent	apparently	rapid	rapidly
Opposite	possibly	impossibly	ethical	unethical
Comparative	great	greater	tough	tougher
Superlative	easy	easiest	lucky	luckiest
Present Participle	think	thinking	read	reading
Nationality adjective	Switzerland	Swiss	Cambodia	Cambodian
Past tense	walking	walked	swimming	swam
Plural nouns	mouse	mice	dollar	dollars
Plural verbs	work	works	speak	speaks

Efficient Estimation of Word Representations in Vector Space

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Table 4: *Comparison of publicly available word vectors on the Semantic-Syntactic Word Relationship test set, and word vectors from our models. Full vocabularies are used.*

Model	Vector Dimensionality	Training words	Accuracy [%]		
			Semantic	Syntactic	Total
Collobert-Weston NNLM	50	660M	9.3	12.3	11.0
Turian NNLM	50	37M	1.4	2.6	2.1
Turian NNLM	200	37M	1.4	2.2	1.8
Mnih NNLM	50	37M	1.8	9.1	5.8
Mnih NNLM	100	37M	3.3	13.2	8.8
Mikolov RNNLM	80	320M	4.9	18.4	12.7
Mikolov RNNLM	640	320M	8.6	36.5	24.6
Huang NNLM	50	990M	13.3	11.6	12.3
Our NNLM	20	6B	12.9	26.4	20.3
Our NNLM	50	6B	27.9	55.8	43.2
Our NNLM	100	6B	34.2	**64.5**	50.8
CBOW	300	783M	15.5	53.1	36.1
Skip-gram	300	783M	**50.0**	55.9	**53.3**

Predictions, for example, *Paris - France + Italy = ? Rome*

Table 8: *Examples of the word pair relationships, using the best word vectors from Table 4 (Skip-gram model trained on 783M words with 300 dimensionality).*

Relationship	Example 1	Example 2	Example 3
France - Paris	Italy: Rome	Japan: Tokyo	Florida: Tallahassee
big - bigger	small: larger	cold: colder	quick: quicker
Miami - Florida	Baltimore: Maryland	Dallas: Texas	Kona: Hawaii
Einstein - scientist	Messi: midfielder	Mozart: violinist	Picasso: painter
Sarkozy - France	Berlusconi: Italy	Merkel: Germany	Koizumi: Japan
copper - Cu	zinc: Zn	gold: Au	uranium: plutonium
Berlusconi - Silvio	Sarkozy: Nicolas	Putin: Medvedev	Obama: Barack
Microsoft - Windows	Google: Android	IBM: Linux	Apple: iPhone
Microsoft - Ballmer	Google: Yahoo	IBM: McNealy	Apple: Jobs
Japan - sushi	Germany: bratwurst	France: tapas	USA: pizza

Efficient Estimation of Word Representations in Vector Space

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Conclusions: "… it is possible to train high quality word vectors using very simple model architectures, compared to the popular neural network models (both feedforward and recurrent). Because of the much lower computational complexity, it is possible to compute very accurate high dimensional word vectors from a much larger data set.

… our comprehensive test set will help the research community to improve …

… high quality word vectors will become an important building block for future NLP applications. "

Google code archive – word2vec
code.google.com/archive/p/word2vec/

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"… The **word2vec** tool takes a text corpus as input and produces the word vectors as output. It first constructs a vocabulary from the training text data and then learns vector representation of words. The resulting word vector file can be used as features in many natural language processing and machine learning applications.

A simple way to investigate the learned representations is to find the closest words for a user-specified word z
The **distance** tool serves that purpose. For example, if you enter 'france', **distance** will display the most similar words and their distances to 'france' …"

Also: large training-set corpora, and pre-trained vectors

Summary

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- M Banko and E Brill. 2001. Scaling to very very large corpora for natural language disambiguation banko01largeCorpora.pdf

  - More data gives higher accuracy, for all classifiers
- T Mikolov et al. 2013. Efficient Estimation of Word Representations in Vector Space mikolov13word2vec.pdf - CBOW, Skip-gram
- Google code archive – word2vec code.google.com/archive/p/word2vec/